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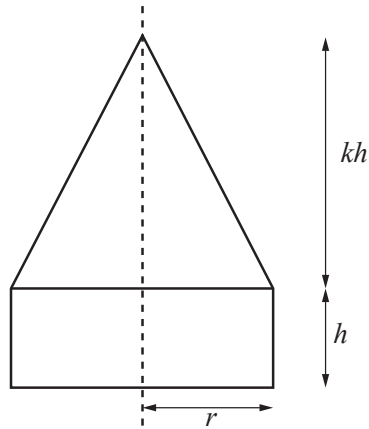
[6]

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An additional particle, of mass $2m$, is now attached to P and the combined particle is released from rest at the original equilibrium position of P . When the combined particle has descended a distance $\frac{1}{3}a$, its speed is $\frac{1}{3}\sqrt{ga}$.

- [illegible]



(a) Show that the distance of the centre of mass of the combined solid from the base of the cylinder is $\frac{h(k^2 + 4k + 6)}{4(3 + k)}$. [4]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

[3]

[illegible]

Find the value of θ and find an expression for u in terms of a and g . [8]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

The total kinetic energy of the spheres after the collision is $\frac{3}{4}mu^2$.

(b) Find the value of α .

[4]

[illegible]

- 7 A particle P is projected from a point O on a horizontal plane and moves freely under gravity. The initial velocity of P is 100 m s^{-1} at an angle θ above the horizontal, where $\tan \theta = \frac{4}{3}$. The two times at which P 's height above the plane is H m differ by 10 s.

(a) Find the value of H .

[5]

- (b)** Find the magnitude and direction of the velocity of P one second before it strikes the plane. [4]

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Additional Page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

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